RW–10927

Sub. Code
511502

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

NUMERICAL METHODS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define :
   (a) Gradient
   (b) Ascent.
2. What is Broyden's method?
3. Define Gauss-Seidal iteration.
4. Define uniform distance.
5. Define least squares approximation.
6. What is trigonometric polynomial and give an example?
7. Define discretization error.
8. Write a corrected trapezoid rule.
9. Define linearly independent.
10. Define :
    (a) Predictor
    (b) Corrector.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Find all critical points of the function $F(x_1, x_2) = x_1^3/3 + x_2^2 x_1 + 3$.

Or

(b) If C is an approximate inverse for the matrix A, i.e. if $\|I - CA\| < 1$ in some matrix norm, prove that both C and A are invertible.

12. (a) Suppose the error $f(x) - p(x)$ in the polynomial approximation $p \in \pi_n$ to f alternates in sign at the points $x_0 \le x_1 < \ldots < x_{n+1}$, i.e., $(-1)^i [f(x_i) - p(x_i)] \in \ge 0$ for $i = 0 \ldots n+1$ with $\in = \text{signum } [f(x_0) - p(x_0)]$. If $a \le x_i \le b$, all i, prove that

$$\text{dis}\, t_\infty(f, \pi_n) \ge \min_i |f(x_i) - p(x_i)|.$$

Or

(b) Find the zeros of the Legendre polynomials $P_2(x)$, $P_3(x)$ and $P_4(x)$.

13. (a) Derive central difference formula for numerical differentiation.

Or

(b) Verify by direct integration that $\int_a^b \psi_2(x)\,dx = 0$ if $\psi_2(x) = (x-a)(x-(a+b)/2)(x-b)$.

14. (a) Expanding $f(a+h)$ and $f(a-h)$ into Taylor series about the point a.

Or

(b) Find the general solution of the difference equations :

(i) $y_{n+1} - 3y_n = 5$

(ii) $y_{n+2} - 4y_{n+1} + 4y_n = n$ (Hint : Find a particular solution, try $y_n^p = an + P = an + b$)

(iii) $y_{n+2} + 2y_{n+1} + 2y_n = 0$

(iv) $y_{n+3} - y_{n+2} + 2y_{n+1} - 2y_n = 0$

(v) $y_{n+2} - y_{n+1} - y_n = 0$.

15. (a) Solve the equation $y' = x^2 + y,\ y(0) = 1$, from $x = 0$ to $x = 0.5$, using Euler's method as a predictor as a corrector. Determine the step h so that four decimal places of accuracy are obtained at $x = 0.5$. Start with $h = 0.05$.

Or

(b) Solve the equation $y' = x + y\ y(0) = 0$ from $x = 1$, using the Adams-Bashforth method.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that Jacobi iteration and Gauss-Seidal iteration converge in finitely many steps when applied to the solution of the linear system $Ax = b$ with A an invertible upper-triangular matrix. Derive the normal equations for the best c_1^*, c_2^* in case $F(x) = F(x : c_1\, c_2) = c_1\, e^{C2x}$ following the argument given in the text. Are these normal equations still linear?

17. Use Simpson's rule to estimate the value of the integral $I = \int_0^1 \left(1 - x^2\right)^{3/2} dx$.

18. Determine an upper bound for the discretization error of Euler's method in solving the equation $y' = y$, $y(0) = 1$ from $x = 0$ to $x = 1$.

19. Solve the following problems, using the shooting method :

    (a) $y'' = 2y^3$, $y(1) = 1$, $y(2) = 1/2$, taking $y'(1) = 0$ as a first guess.

    (b) $y'' = e^y$, $y(0) = y(1) = 0$, taking $y'(0) = 0$ as a first guess.

20. The determination of the condensation of a pure vapor on the outside of a cooled horizontal tube requires that the mean heat-transfer coefficient Q be computed. This coefficient requires, along with other parameters, the evaluation of the integral $I = \int_0^{\pi} (\sin x)^{\frac{1}{3}} dx$. Find the value of this integral using Simpson's rule with $N = 5, 10, 15, 20$ subdivisions.

RW–10887

Sub. Code
511202

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Mathematics

MEASURE AND INTEGRATION

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a measurable set.
2. Show that every countable set has measure zero.
3. Define a measurable function.
4. Show that continuous functions are measurable.
5. Show that if f is integrable then f is finite-valued $a.e$.
6. Show that if f and g are measurable, $|f| \leq |g|$ $a.e$ and g is integrable, then f is integrable.
7. Define a signed measure.
8. Show that a countable union of sets positive with respect to a singed measure ν s a positive set.
9. Show that if $\{A_i\}$ is a monotone sequence of sets, then $\lim A_i^y - (\lim A_i)^y$ and $\lim(A_i)_x = (\lim A_i)_x$.
10. Let A be a non measurable set in x. Is $A x \phi$ a measurable rectangle?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Show that an outer measure is translation invariant.

Or

(b) Let μ^* be the outer measure on $\mathcal{H}(R)$ defined by μ on R then show that S^* contains $S(R)$, the σ-ring generated by R.

12. (a) Prove that not every measurable set is a Barel set.

Or

(b) Let f and g be real-valued measurable functions defines on the same measurable set. Then show that $f+g$, $f-g$ and fg are also measurable.

13. (a) Let f and g be nonnegative measurable functions. Then show that $\int f dx + \int g dx = \int (f+g) dx$.

Or

(b) State and prove Lebesgue's dominated convergence theorem.

14. (a) Show that if v_1, v_2 and μ are measures and $v_1 \perp \mu_1$, $v_2 \perp \mu$ then $v_1 + v_2 \perp \mu$.

Or

(b) Let v be a singed measure on $[\![X, S]\!]$. Let $E \in S$ and $v(E) > 0$. Show that there exists A, a set positive with respect to v, such that $A \subseteq E$ and $v(A) > 0$.

15. (a) If $E \in S \times \mathcal{T}$, then show that for each $x \in X$ and $y \in Y$. $E_x \in \mathcal{T}$ and $E^y \in S$.

Or

(b) Show that if f is S-measurable and g is $\mathcal{T}$ measurable then show that fg is $S \times \mathcal{T}$ measurable.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Show that the outer measure of an interval equals its length.

17. Show that there exists a non-measurable set.

18. State and prove Fatous' lemma and hence deduce Lebesgue's monotone convergence theorem.

19. State and prove Radon-Nikodym theorem.

20. State and prove Fubini's theorem.

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RW–10888

Sub. Code
511203

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Mathematics

PARTIAL DIFFERENTIAL EQUATIONS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Show whether the equation $z(z+y)\,dx + z(z+x)\,dy - 2xy\,dz = 0$ is integrable.

2. Solve the equation $(x^2z - y^3)\,dx + 3xy^2dy + x^3dx = 0$.

3. Form a partial differential equation by eliminating the arbitrary function f from the equation $z = f\left(\frac{xy}{z}\right)$.

4. Define complete integral and general solution of a partial differential equation.

5. Find a complete integral of the equation $pq = 1$.

6. Find a complete integral of the equation $(p+q)(z - xp - yq) = 1$.

7. If $u = f(x+iy) + g(x-iy)$ where f and g are arbitrary functions, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

8. Solve the equation $\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial y^4} = 2\frac{\partial^4 z}{\partial x^2 \partial y^2}$.

9. Define Dirichlet and Neumann boundary value problem for Laplace equation.

10. State the elementary solution of a Laplace equation.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Find the integral curves of the equations $\frac{dx}{x+z} = \frac{dy}{y} = \frac{dz}{z+y^2}$.

Or

(b) Solve the equation

$yz(y+z)\,dx + xz(x+z)\,dy + xy(x+y)\,dz = 0$.

12. (a) Find the general solution of the equation $x^2\frac{\partial z}{\partial x} + y^2\frac{\partial z}{\partial y} = (x+y)z$.

Or

(b) Find the characteristics of the equation $pq = z$.

13. (a) Find the complete integral of the equation $(p^2+q^2)y = qz$.

Or

(b) Show that the equations $xp - yq = x, x^2p + q = xz$ are compatible and find their solution.

14. (a) Find the solution of the equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x - y$.

Or

(b) Reduce the equation $\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$ to canonical form and hence solve it.

15. (a) Prove that the solutions of a certain Neumann problem can differ from one another by a constant only.

Or

(b) State and derive the d'Alembert's solution of the one dimensional wave equation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the orthogonal trajectories on the conicoid $(x+y)z=1$ of the conics in which it is cut by the system of planes $x-y+z=k$ where k is a parameter.

17. Find the integral surface of the equation $x(y^2+z)p-y(x^2+z)q=(x^2-y^2)z$ which contains the line $x+y=0, z=1$.

18. Find a complete integral of the equation $p^2x+qy=z$ and hence derive the equation of an integral surface of which the line $y=1,\ x+z=0$ is a generator.

19. Determine the solution of the equation $\frac{\partial^2 V}{\partial r^2}+\frac{1}{r}\frac{\partial V}{\partial r}+\frac{\partial^2 V}{\partial z^2}=0$ for the region $r \geq 0$ as $z \geq 0$, satisfying the conditions.

(a) $V \to \infty$ as $z \to \infty$ and as $r \to \infty$

(b) $V=f(r)$ on $z=0, r \geq 0$.

20. Find the temperature in a sphere of radius a when its surface is maintained at zero temperature and its initial temperature is $f(r,\theta)$.

RW–10889

Sub. Code
511204

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Mathematics

COMPLEX ANALYSIS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Prove that an analytic function with constant modulus is constant.

2. Expand $(1-z)^{-m}, m$, a positive integer, in powers of z.

3. Evaluate $\oint_C \frac{z^2+5}{z-3} dz$, where C is the circle $|z| = 4$.

4. State the Cauchy integral formula for the higher derivative.

5. Define essential singularity with an example.

6. Discuss singularity of e^z at $z = \infty$.

7. State the Argument principle.

8. Find the poles and residues of $\cot z$.

9. State Weierstrass's theorem.

10. If $f(z)$ is of genus h, how large and how small can the genus of $f(z^2)$ be?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Prove that every rational function has a partial fraction expansion.

Or

(b) Construct an analytic function whose real part is $u(x, y) = x^3 - 3xy^2 + y$.

12. (a) If $pdx + qdy$ is locally exact in Ω, then prove that $\int_\gamma pdx + qdy = 0$

Or

(b) If $f(z)$ is analytic on a closed curve y then show that $\int_\gamma \overline{f(z)f'(z)dz}$.

13. (a) Show that any function which is meromorphic in the extended plane is rational.

Or

(b) State and prove maximum principle theorem.

14. (a) Evaluate $\int_0^\infty \frac{x^2 dx}{x^2 + 5x^2 + 6}$ by the method of residues.

Or

(b) If u_1 and u_2 are harmonic in region Ω, then prove that $\int_\gamma u_1^* d\, u_2 = u_2^* d\ u_1 = 0$ for every cycle γ which is homologous to zero in Ω.

15. (a) Show that $\prod_{n=2}^{\infty}\left(1 - \frac{1}{n^2}\right) = \frac{1}{2}$.

Or

(b) Develop $\log\left(\frac{\sin z}{z}\right)$ in powers of z up to the term z^6.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) If $R(z)$ is a rational function of order n, how large and how small can the order of $R'(z)$ be?

(b) State and prove Abel's limit theorem.

17. Suppose $\Phi(\zeta)$ is continuous on a curve γ in the finite complex plane. Then $F_n(z) = \int \frac{\Phi(\zeta)}{(\zeta - z)^n} d\zeta \; (n = 1,2,3,....)$ is defined and analytic in each of the regions determined by γ and $F_n(z) = n F_{n\to 1}(z)$.

18. (a) Prove that a nonconstant analytic function maps open sets onto open sets.

(b) Prove that a region Ω is simply connected if and only if $\eta(\gamma, a) = 0$ for all cycles γ in Ω and all points α which do not belong to Ω.

19. (a) State and prove the Cauchy's residue theorem.

(b) If u is harmonic and bounded in $0 < |z| < \rho$, show that the origin is a removable singularity in the sense that u becomes harmonic in $|z| < \rho$ when $u(0)$ is properly defined.

20. (a) Express the Taylor development of $\tan z$ and Laurent development of cot z in terms of the Bernoulli numbers.

(b) Prove that the value of an absolutely convergent product does not change if the factors are reordered.

RW–10890

Sub. Code
511401

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Mathematics

PROBABILITY AND STATISTICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. If C_1 and C_2 are subsets of $\mathcal{C}$ such that $C_1 \subset C_2$ then prove that $p(C_1) \le p(C_2)$.

2. Define random variable.

3. The p.d.f of two random variables x and y, is $f(x,y) = \begin{cases} 6x^2y & if\ 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$

   variables find $\Pr(0 < x < 3/4, 1/3 < y < 2)$.

4. Prove that $E[(x-\mu_1)(y-\mu_2)] = E(xy) - \mu_1\mu_2$.

5. Define Normal distribution.

6. Define Poisson distribution.

7. State some properties of t distribution.

8. Let X have the p.d.f $f(x) = 1/3 x = 1,2$, zero elsewhere. Find p.d.f of $Y = 2X + 1$.

9. Let $\overline{X}$ be the mean of a random sample of size 25 from a distribution that is $N(75,100)$. Find $\Pr(71 < \overline{X} < 79)$.

10. Define converges in distribution.

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) If C_1 and C_2 are subsets of $\mathcal{C}$, then prove that $P(C_1 \cup C_2) =)P(C_1) + P(C_2) - P(C_1 \cap C_2)$.

Or

(b) For given distribution function

$$f(x) = \begin{cases} 0 & x < 1 \\ \dfrac{x+2}{4} & -1 \le x < 1 \\ 1 & 1 \le x \end{cases}$$

Find $\Pr(x = 0), \Pr(x = 1), \Pr(2 < x \le 3)$.

12. (a) Let $f(x_1, x_2) = 4x_1x_2 0 < x_1 < 1, 0 < x_2 < 1$, zero elsewhere be the p.d.f of x_1 and x_2 find

(i) $\Pr(0 < x_1 < \frac{1}{2}, \frac{1}{4} < x_2 < 1)$

(ii) $\Pr(x_1 = x_2)$,

(iii) $\Pr(x_1 \le x_2)$.

Or

(b) X_1 and X_2 have the p.d.f

$$f(x_1, x_2) = \begin{cases} 8x_1x_2 & 0 < x_1 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find $E(7x_1x_2^{\ 2} + 5x_2)$.

13. (a) Let X be the number of heads in n = 7 independent tosses of an unbiased coin. p.d.f of x is

$$f(x) = \begin{cases} \binom{7}{x} \left(\frac{1}{2}\right)^X \left(1 - \frac{1}{2}\right) & \text{x = 0,1...7} \\ 0 & elsewhere \end{cases}$$

Find $\Pr(0 < x < 1), \Pr(x = 5)$.

Or

(b) Let X have a gamma distribution with $\alpha = r/2$ where r is a positive integer, and $\beta > 0$. Define the random variable $Y = 2x/\beta$.

14. (a) Let Y_1, Y_2, Y_3 be the order statistics of a random sample of size 3 from a distribution having p.d.f $f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$ find the p.d.f of the sample range $Z_1 = Y_3 - Y_1$.

Or

(b) If $X_1, X_2 ... X_n$ are independent random variable with respect to moment generating functins $\mu_i(t) i = 1,...n$ then prove that moment generating function of $Y = \sum_{i=1}^{n} a_i x_1$, where $a_1, a_2 .. a_k$ are real constants, is $\mu_\gamma(t) = \prod_{i=1}^{n} \mu_i(a_i t)$.

15. (a) If $\overline{X}$ is the mean of a random sample of size n from a normal distribution with mean μ and variance 100, show that $\Pr(\mu - 5 < \overline{X} < \mu + 5) = 0.954$.

Or

(b) Let $F_n(u)$ denote the distribution function of a random variable U_n whose distribution depends upon the positive integer n, let U_n converges in probability to the constant $c \neq 0$. Prove that the random variable $U_{n/c}$ converges in probability 1.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove Chebyshev's inequality.

17. Prove that

    (a) $E[E(X_2 / X_1)] = E(X_2)$

    (b) $\text{var}[E(X_2 / X_1)] \leq \text{var}\, X_2$.

18. Compute the measures of skewness and kurtosis of the binomial distributin $B(n, p)$.

19. Derive the beta distribution.

20. Let $X_1, X_2 ... X_n$ denote the observation of a random sample from a distribution with has mean positive variable σ^2 then prove that the random variable $Y_n = (\sum_i^n X_i - n\mu) / \sqrt{n\sigma} = \sqrt{n}(\overline{X}_n - \mu) / \sigma$ has a limiting distribution that is normal with mean zero and variance 1.

RW–10891

Sub. Code
511402

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Mathematics

FUNCTIONAL ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Define Hamel basis of a vector space with an example.

2. Define normed linear space with an example.

3. Define Hilbert space with an example.

4. Give examples of subspaces of l^2 ?

5. Define Hibert-adjoint operator.

6. Show that $0^* = 0$.

7. Define adjoint operator.

8. Define Canonical mapping with an example.

9. Define strong convergence.

10. Define open map with an example.

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) State and prove the Riesz's Lemma.

Or

(b) Show that $\mathbb{R}^n$ is not compact.

12. (a) If in an inner product space, $< x,u >=< x,v) >$ for all x, show that $u = v$.

Or

(b) State and prove Schwarz inequality.

13. (a) Show that the dual space of the real space l^2 is l^2.

Or

(b) Show that the column vectors of a unitary matrix constitute an orthonormal set with respect to the inner product on $\mathbb{C}^n$.

14. (a) Show that a norm on a vector space X is a sublinear functional on X.

Or

(b) What are the adjoints of a zero operator 0 and an identity operator I?

15. (a) Prove that strong convergence implies weak convergence with the same limit.

Or

(b) Show that an open mapping need not map closed sets onto closed sets.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Let X be the vector space of all complex 2 × 2 matrices and define $T : X \to X$ by $Tx = bx$, where $b \in X$ is fixed and bx denotes the usual product of matrices. Show that T is linear. Under what condition does T^{-1} exist?

17. Show that in an inner product space, $x \perp y$ if and only if $\|x + \alpha y\| \geq \|x\|$ all scalars α.

18. State and prove the Riesz's Theorem.

19. State and prove the Hahn-Banach Theorem.

20. State and prove the closed graph theorem.

RW–10892

Sub. Code
511504

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Mathematics

OPTIMIZATION TECHNIQUES

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define shortest- route problem.
2. Define critical and non critical.
3. Define convex set.
4. Define extreme point.
5. Explain two person zero sum game.
6. Define optimal strategy.
7. State Kuhn- Tucker conditions.
8. Explain lagrangian method.
9. Define separable function with an example.
10. Define Excel moment.

Section B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write the algorithm for minimal spanning tree.

Or

(b) Write the Dijkstra's algorithm.

12. (a) Determine and classify all the basic solution of the following system of equation

$$\begin{pmatrix} 1 & 3 & -1 \\ 2 & -2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

Or

(b) Write the algorithm for Revised simplex method.

13. (a) Find the range of values of p and q which will render the entry (2,2) a saddle point for the game.

Player B

Player A	B_1	B_2	B_3
A_1	2	4	5
A_2	10	7	q
A_3	4	p	6

Or

(b) Player A and B each take out one (or) two matches and guess how many matches the opponent has taken. If one of the player guesses correctly the loser has to pay him as many rupees as the sum of the number held by both players. Otherwise the payout is zero. Write down the payoff matrix and obtain the optimal strategies of both players.

14. (a) Prove that a necessary condition for x_0 to be an extreme point of $f(x)$ is that $\nabla(f(x_0))=0$.

Or

(b) Prove that the sufficiency of the Karush - Kuhn Tucker conditions.

15. (a) Maximize $f(x)=\begin{cases} 3x & 0\le x\le 2 \\ \frac{1}{3}(-x+20) & 2\le x\le 3 \end{cases}$

The maximum value of $f(x)$ occurs at $x = 2$. Demonstrate the calculations for iterations 1 and 2 by using golden section method $(\text{Assume } \Delta=0.1)$.

Or

(b) Explain Gradient method.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write the algorithm for maximal flow.

17. Solve the following LPP model by using the upper bounding algorithm

Maximize $z=3x_1+5y+2x_3$

Subject to $x_1+y+2x_3\le 14$

$2x_1+4y+3x_3\le 43$

$0\le x_1\le 4,\ 7\le y\le 10,\ 0\le x_3\le 3.$

The lower bound on y is recounted for using the substitution $y=x_2+7$ where $0\le x\le 3$.

18. For the following pay off matrix transform the zero sum game into an equivalent linear programming problem and solve it by using simplex method.

	Player Y		
Player X	Y_1	Y_2	Y_3
X_1	1	–1	3
X_2	3	5	–3
X_3	6	2	–2

19. Find the derivative of the Newton - Raphson method.

20. Maximize $z = x_1 - x_2$

Subject to $3x_1^{\ 4} + x_2 \leq 243$

$x_1 + 2x_2^2 \leq 32$, $_1 \geq 2.1$, $x_2 \geq 3.5$

Find the solution by using separable convex programming.

RW–10910

Sub. Code
511101

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

ALGEBRA – I

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Give an example for abelian group of order 2.

2. Define the order of an element $'a'$ in a group G.

3. Define the kernel of homomorphism of $\phi : G \to G'$.

4. If $\theta = (1, 3, 2)(4)$ and $\psi = (1)(2, 3)(4)$ find $\psi_o \theta$.

5. Define a p-Sylow subgroup of G.

6. Explain the internal direct product of normal subgroups $N_1, N_2, ..., N_n$ in a group G.

7. Define a division ring.

8. Write an example for zero homomorphism.

9. If an ideal U of a ring contains a unit of R then _____.

10. What do you mean by domain of Gaussian Integers?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Prove that H is a sub group of a group G if H is a non empty finite subset of G and is closed under Multiplication.

Or

(b) Prove that the subgroup N of G is normal subgroup of G if and only if every left coset of N in G is right coset of N in G.

12. (a) If ϕ is a homomorphism of G into $\overline{G}$ with kernel K then prove that K is a normal subgroup of G.

Or

(b) If G is a group and ϕ an automorphism of G, if $o(a) > 0$ then prove that $o(\phi(a)) = o(a)$.

13. (a) State and prove the first part of Sylow's theorem.

Or

(b) If G and G' are isomorphic abelian groups then prove that for every integers, $G(s)$ and $G'(s)$ are isomorphic.

14. (a) If U is an ideal of the ring R, prove that $\frac{R}{U}$ is a ring and is a homomorphic image of R.

Or

(b) If R is a commutative ring with unit element and M is an ideal of R prove that M is a maximal ideal of R iff $\frac{R}{M}$ is a field.

15. (a) Prove that Euclidean ring possesses a unit element.

Or

(b) If $f(x)$ and $g(x)$ are primitive polynomials, then prove that $f(x)g(x)$ is a primitive polynomial.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. If H and K are finite subgroups of G of order $o(H)$ and $o(K)$ then prove that $o(HK) = \frac{o(H)o(K)}{o(H \cap K)}$.

17. State and prove the class equation of a finite group G.

18. Prove that every integral domain can be imbedded in a field.

19. State and prove Unique Factorization theorem.

20. Prove that $J[i]$ is a Euclidean ring.

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RW–10911

Sub. Code
511102

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

REAL ANALYSIS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Give an example of a set which has no greatest lower bound.

2. Define compact of a metric space.

3. Define the uniformly continuous function on a metric space.

4. Give an example for f has discontinuity of the second kind at every point x.

5. Prove that f is continuous at x if f is differentiable at a point $x \in [a, b]$.

6. Define a local maximum for a real valued function.

7. Explain the upper and lower Riemann integrals on $[a, b]$.

8. Define the unit step function.

9. Define the uniformly convergent sequence.

10. Write an example for uniformly bounded function on $[0, 1]$.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Prove that if $x \in R, y \in R$, and $x < y$ then there exists a $p \in Q$ such that $x < p < y$.

Or

(b) Prove that a set E is open if and only if its complement is closed.

12. (a) Prove that a mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y.

Or

(b) Let f be monotonic on (a, b). Show that the set of points on (a, b) at which f is discontinuous is at most countable.

13. (a) Give an example of a function $f(x)$ which is differentiable at all points x but the first derivative is not a continuous function.

Or

(b) Suppose f is a continuous mapping of $[a, b]$ into R^k and f is differentiable in (a, b) show that there exists $x \in (a, b)$ such that $|f(b) - f(a)| \leq (b - a)|f'(x)|$.

14. (a) If P^* is a refinement of P prove that $L(P, f, \alpha) \leq (P^*, f, \alpha)$ and $U(P^*, f, \alpha) \leq U(P, f, \alpha)$.

Or

(b) If f maps $[a, b]$ into $\mathbb{R}$ and if $f \varepsilon R(\alpha)$..... for some monotonically increasing function α on [a, b] then prove that $|f| \varepsilon R(\alpha)$.... and $\left|\int_a^b f\, d\alpha\right| \leq \int_a^b |f|\, d\alpha$.

15. (a) Prove that the sequence of function $\{f_n\}$ converges uniformly on E if and only if for every $\varepsilon > 0$ other exists an integer N such that $m \geq N, n \geq N, x \in E$ then prove that $|f_n(x) - f_m(x)| \leq \varepsilon$.

Or

(b) Give an example of a convergent sequence need not contain a uniformly convergent subsequence.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that every k-cell is compact.

17. Let f be a continuous mapping of a compact metric space X into a metric space Y then prove that f is uniformly continuous on X.

18. State and prove Taylor's theorem.

19. Suppose $f \in R(\alpha)$ on $[a, b]$, $m \leq f \leq M, \Phi$ is continuous on $[m, M]$ and $h(x) = \Phi(f(x))$ on $[a, b]$. Then prove that $h \in R(\alpha)$ on $[a, b]$.

20. State and prove the Stone-Weierstrass theorem.

RW–10912

Sub. Code
511103

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

ORDINARY DIFFERENTIAL EQUATIONS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

Each question carries 2 marks.

1. Write the Wronskian function of ϕ_1, ϕ_2.

2. Show that the functions $\phi_1(x) = x^2, \phi_2(x) = x|x|$ for $-\infty < x < \infty$ are linearly independent.

3. Two solutions of $x^3y''' - 3xy' + 3y = 0, (x > 0)$, are $\phi_1(x) = x, \phi_2(x) = x^3$, find a third independent solution.

4. Define Legendre's equation.

5. Find all solutions of the following equation for $x > 0$: $x^2y'' + 2xy' - 6y = 0$.

6. Obtain two linearly independent solutions of the following equation which are valid near $x = 0$: $x^2 y'' + 3xy' + (1 + x)y = 0$.

7. Give Lipschitz condition.

8. Write a note about the concept of successive approximation.

9. Consider the problem $y' = y + \lambda x^2 \sin y, y(0) = 1$ where λ is some real parameter, $|\lambda| \le 1$. Prove that $\left|\psi(x) - e^x\right| \le |\lambda|\left(e^{|x|} - 1\right)$ for $|x| \le 1$.

10. What do you mean by non – local existence of solutions?

Part B (5 × 5 = 25)

Answer **all** the questions by choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Let $r_1, \cdots, r_s$ be the distinct roots of the characteristic polynomial p and suppose r_i has multiplicity m_i (thus $m_1 + m_2 + \cdots + m_s = n$). Then prove that the n function

$e^{r_1 x}, xe^{r_1 x}, \cdots, x^{m_1 - 1} e^{r_1 x}; e^{r_2 x}, xe^{r_2 x}, \cdots, x^{m_2 - 1} e^{r_2 x}; \cdots; e^{r_s x}$, $xe^{r_s x}, \cdots, x^{m_s - 1} e^{r_s x}$ of $L(y) = 0$ are solutions of $L(y) = 0$.

Or

(b) Let $\alpha_1,...,\alpha_n$ be any n constants and let x_0 be any real number. Then prove that there exists a solution ϕ of $L(y)=0$ on $-\infty < x < \infty$ satisfying $\phi(x_0)=\alpha_1, \phi'(x_0)=\alpha_1,...,\phi^{(n-1)}(x_0)=\alpha_n$.

12. (a) If ϕ_1 is a solution of $\overline{L}(y)=y''+a_1(x)y'+a_2(x)y=0$ on an interval I and $\phi_1(x)\neq 0$ on I, a second ϕ_2 of $\overline{L}(y)$ on I is given by $\phi_2(x)=\phi_1(x)\int_{x_0}^{x}\frac{1}{[\phi_1(s)]^2}\exp\left[-\int_{x_0}^{s}a_1(t)\,dt\right]ds$. Then prove that the functions ϕ_1,ϕ_2 form a basis for the solution of $\overline{L}(y)$ on I.

Or

(b) Show that $P_n(-x)=(-1)^n P_n(x)$, and hence prove that $P_n(-1)=(-1)^n$.

13. (a) Find the singular points of the following equation and determine those which are regular singular points : $3x^2y''+x^6y'+2xy=0$.

Or

(b) Suppose ϕ is any solution of $x^2y''+xy'+x^2y=0$ for $x>0$, and let $\psi(x)=x^{1/2}\phi(x)$. Show that ψ satisfies the equation $x^2y''+\left(x^2+\frac{1}{4}\right)y=0$ for $x>0$.

14. (a) Find all real – valued solutions of the equation : $y' = x^2 y$.

Or

(b) Consider the initial value problem $y' = 3y + 1,\ y(0) = 2$. Compute the first four approximations $\phi_0, \phi_1, \phi_2, \phi_3$ to the solution.

15. (a) Let f be continuous and satisfy a Lipschitz condition on R. If ϕ and ψ are two solutions of $y' = f(x, y), y(x_0) = y_0$, on an interval I containing x_0, then prove that $\phi(x) = \psi(x)$ for x in I.

Or

(b) Let f be a real – value continuous function on the strip $S : |x| \le a, |y| < \infty$, $(a > 0)$ and suppose f satisfies a Lipschitz condition on S. Show that the solution of the initial value problem $y'' + \lambda^2 y = f(x, y),\ y(0) = 0,\ y'(0) = 1$, $(\lambda > 0)$ is unique.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. Let ϕ be any solution of $L(y) = y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0$ on an interval I containing a point x_0. Then prove that for all x in I, $\|\phi(x_0)\| e^{-k|x - x_0|} \le \|\phi(x)\| \le \|\phi(x_0)\| e^{k|x - x_0|}$ where $k = 1 + |a_1| + \cdots + |a_n|$

17. Let b be continuous on an interval I and let $\phi_1,\cdots,\phi_n$ be a basis for the solutions of $L(y)=0$ on I. Every solution ψ of $L(y)=b(x)$ can be written as $\psi=\psi_p+c_1\phi_1+\cdots+c_n\phi_n$, where ψ_p is a particular solution $L(y)=b(x)$ and $c_1,\cdots,c_n$ are constants. Every such ψ is a solution $L(y)=b(x)$. Prove that a particular solution ψ_p is given by $\psi_p(x)=\sum_{k=1}^{n}\phi_k(x)\int_{x_0}^{x}\frac{W_k(t)b(t)\,dt}{W(\phi_1,\cdots,\phi_n)(t)}$.

18. Find all solutions ϕ of the form $\phi(x)=|x|^r\sum_{k=0}^{\infty}c_kx^k,(|x|>0)$, for the following equation : $x^2y''+xy'+x^2y=0$.

19. Prove that a function ϕ is a solution of the initial value problem $y'=f(x,y)$, $y(x_0)=y_0$ on an interval I if and only if it is a solution of the integral equation $y=y_0+\int_{x_0}^{x}f(t,y)dt$ on I.

20. Let f,g be two continuous vector real valued function on $R:|x-x_0|\le a$, $|y-y_0|\le b,(a,b>0)$ and suppose f satisfied a Lipschitz condition on R with Lipschitz constant K. Suppose ϕ,ψ are solutions of the problems $y'=f(x,y)$, $y(x_0)=y_1$ and $y'=g(x,y)$, $y(x_0)=y_2$

respectively, on some interval I containing x_0. If for $\in, \delta \geq 0, |f(x,y) - g(x,y)| \leq \in$, (all (x,y) in R) and $|y_1 - y_2| \leq \delta$, then prove that $|\phi(x) - \psi(x)| \leq \delta e^{K|x-x_0|} + \frac{\in}{K}\left(e^{K|x-x_0|} - 1\right)$ for all x in I. Also prove that $y' = f(x,y)$, $y(x_0) = y_0$ has at most one solution on any interval containing x_0.

RW–10913

Sub. Code
511104

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

ANALYTIC NUMBER THEORY

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. Prove that there are infinitely many prime numbers.
2. Define Euler totient function.
3. State Dirichlet's formula.
4. Prove that $[2x]+[2y] \geq [x]+[y]+[x+y]$.
5. Define Chebyshev's ψ -function.
6. State Mertens conjecture.
7. If $ac \equiv bc(\bmod m)$ and if $d=(m,c)$, then prove that $a \equiv b\left(\bmod \frac{m}{d}\right)$.
8. Find all n for which $\phi(n) = 2(\bmod 4)$.
9. Find $(22|11)$.
10. Define Jacobi symbol.

Part B (5 × 5 = 25)

Answer **all** questions by choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Prove that every integer $n > 1$ is either a prime number or a product of prime numbers.

Or

(b) If $n \geq 1$ then prove that $\sum_{d|n} \phi(d) = n$.

12. (a) For all $x \geq 1$ then prove that $\sum_{n \leq x} \sigma_1(n) = \frac{1}{2}\varsigma(2)x^2 + 0(x \log x)$.

Or

(b) State and prove Legendre's identity.

13. (a) For $x > 0$ then prove that $0 \leq \frac{\psi(x)}{x} - \frac{\theta(x)}{x} \leq \frac{(\log x)^2}{2\sqrt{x}\log^2}$.

Or

(b) For every integer $n \geq 2$ then prove that $\frac{1}{6}\frac{n}{\log n} < \pi(n) < 6\frac{n}{\log n}$.

14. (a) Assume $(a, m) = 1$. Then Prove that the linear congruence $ax \equiv b(\text{mod}\, m)$ has exactly one solution.

Or

(b) If $(a, m) = 1$ then prove that the solution (unique mod m) of the linear congruence $ax \equiv b(\text{mod}\, m)$ is given by $x \equiv ba^{\phi(m)-1}(\text{mod}\, m)$.

15. (a) Prove that the Legendre's symbol is a completely multiplicative function.

Or

(b) For every odd prime p, prove that

$$(2/p) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & if\ p \equiv \pm 1 \pmod 8 \\ -1 & if\ p \equiv \pm 3 \pmod 8 \end{cases}$$

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. State and prove Euclidean algorithm.

17. For $x \geq 2$ prove that $\sum_{p \leq x} \left[\frac{x}{p}\right] \log p = x \log x + 0(x)$ where the sum is extended over all primes $\leq x$.

18. For all $n \geq 1$ prove that nth prime p_n satisfied the inequalities $\frac{1}{6} n \log n < p_n < 12\left(n \log n + n \log \frac{12}{e}\right)$.

19. For a prime p, let $f(x) = c_0 + c_1 x + \ldots + c_n x^n$ be a polynomial of degree n with integer coefficients such that $c_n \not\equiv 0 \pmod p$. Then prove that the polynomial congruence $f(x) \equiv 0 \pmod p$ has at most n solutions.

20. If p and q are distinct odd primes then prove that $(p|q)(q|p) = (-1)^{(p-1)(q-1)/4}$.

RW–10914

Sub. Code
511513

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

OBJECT ORIENTED PROGRAMMING AND C++

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. List the Rules for naming the identifiers in C++.

2. What are manipulators?

3. What is the use of friend function?

4. What is Constructor?

5. What is the use of copy constructor?

6. What is new( ) operator?

7. List out the Operators that cannot be overloaded.

8. Define: Polymorphism.

9. What is the use of inheritance?

10. Define: Virtual Function.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Briefly explain about Inheritance.

Or

(b) Briefly explain about Message Passing.

12. (a) Write a C++ program to find the sum of two numbers.

Or

(b) (i) How the class is specified?

(ii) How to create an object?

13. (a) List the special characteristics of constructor.

Or

(b) Explain any five string functions with example.

14. (a) With suitable example explain the function overloading.

Or

(b) Explain operator overloading with an example.

15. (a) Write a C++ program to illustrate virtual function.

Or

(b) With suitable example, explain run-time polymorphism.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the basic concept of Object oriented programming.

17. With suitable example explain the following :

    (a) friend function

    (b) inline function.

18. Write a C++ program to arrange the names in ascending order.

19. An electricity board charges the following rates to domestic users :

    For the first 100 units — 40p/unit

    For the next 200 units — 50p/unit

    Beyond 300 units — 60p/unit.

    Write a C++ program to read the names of users and number of units consumed and print out the charges with names.

20. With suitable example, explain static and dynamic binding.

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RW–10915

Sub. Code
511301

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

MECHANICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define equations of motion For a system of N particle.

2. Write the principle of conservation of energy.

3. Write the standard form of Lagrange's equations for a holonomic system.

4. Define Routhian function.

5. Write the first variation of the function f at the stationary point q_0.

6. Write the Jacobi's form of the principle of least action.

7. Define the solution of the Hamilton problem.

8. Write the Hamilton's equation.

9. Write the new Lagrangian function.

10. Give an example for the identity transformation of a canonical transformation.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write a short note on Generalized coordinates.

Or

(b) Discuss the kinetic energy of a system.

12. (a) Explain Integrals of the motion.

Or

(b) Derive the Jacobi integral.

13. (a) Find the stationary values of the functions $f = z$ subject to the constraints

$\phi = x^2 + y^2 + z^2 - 4 = 0$

$\phi_2 = xy - 1 = 0$

Or

(b) Explain Geodesic problem.

14. (a) State and prove Jacobi's theorem.

Or

(b) Illustrate the Hamilton – Jacobi method, for simple mass – spring system.

15. (a) Show that the transformation

$\theta = \sqrt{2q}\ e^t \cos p$, $p = \sqrt{2q}\ e^{-t}\ \sin p$ is canonical.

Or

(b) Write a short note on Homogeneous canonical transformation.

Part C (3× 10 = 30)

Answer any **three** questions.

16. Discuss the following (a) Angular momentum and (b) Generalized momentum.

17. (a) Derive the Lagrange's equation

    (b) Find the differential equations of motion for a spherical pendulum.

18. Derive the Hamilton's canonical equations of motion.

19. Explain Hamilton's principal function.

20. Discuss on Poisson brackets, and show that the transformation $Q = \sqrt{e^{-2q} - p^2}$, $P = \cos^{-1}\left(pe^{q}\right)$ is canonical.

RW–10916

Sub. Code
511302

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

Topology

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. What is the maximum number of open sets in a topological space X having n points?

2. Let X be any set. Give a basis for the indiscrete topology on X.

3. Define $d((x_1, y_1), (x_2, y_2)) = |x_1 - y_1| - |x_2 - y_2|$ on $\mathbb{R}^2 \times \mathbb{R}^2$. Verify whether d is a metric on $\mathbb{R}^2$ or not.

4. Is $\mathbb{R}$ connected under order topology? Justify your answer.

5. Prove that the image of a compact set under a continuous function is compact.

6. Give an example of a set which is connected but not compact. Justify your answer.

7. Give an example of a space which is not first countable.

8. Give an example of a Hausdorff space which is not regular.

9. Define compactification.

10. Give any two compactifications for $\mathbb{R}$.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Let Y be a subspace of a topological space X. Prove that a subset A is closed in Y if and only if it is the intersection of some closed subset of X with Y.

Or

(b) (i) Define order topology in an ordered set.

(ii) Prove that it is indeed a topology.

12. (a) Prove that the image of a connected set under a continuous map is connected.

Or

(b) Let $p : X \to Y$ be a continuous map. If there exists a Continuous map $f : Y \to X$ such that p o f equals the identity map of Y, then show that p is a quotient map.

13. (a) Prove that every compact subset of a Hausdorff space is closed.

Or

(b) Prove that the compactness implies the limit point compactness but not conversely

14. (a) If X is second countable then prove that every open Covering of X contains a countable subcollection covering X.

Or

(b) Prove that the product of any family of regular spaces is regular

15. (a) Define completely regular space, give an example of a completely regular space and prove your claim.

Or

(b) Show that a connected normal space having more than two points is uncountable.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

Each question carries 10 marks.

16. Let $U \subseteq X$. If $\Im$ and $\Im'$ are topologies on X, and $\Im'$ is strictly finer than $\Im$, then what can you say about the corresponding subspace topologies on the subset Y of X? Justify your answer.

17. (a) State and prove uniform limit theorem.

(b) Prove that the set $\{(x, y) / xy = 2\}$ is a closed subset of $\mathbb{R}^2$.

18. State and prove the classical version of Ascoli's theorem.

19. Prove the following:

    (a) The product of Lindelöf spaces need not be Lindelöf.

    (b) A subspace of Lindelöf space need not be Lindelöf.

20. State and prove Tychonoff theorem.

RW–10917

Sub. Code
511303

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

DIFFERENTIAL GEOMETRY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define functions of class m.

2. Define the curvature.

3. Write a short note on three point contact.

4. Define osculating sphere.

5. Write the parametric equations of a surface.

6. Define normal component.

7. Write a characteristic property of a geodesic.

8. Write the Christoffel symbols of the first kind.

9. Define second fundamental coefficients.

10. Write a necessary and sufficient condition for a surface to be a developable.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Prove that a necessary and sufficient condition that a curve be a straight line is that $\alpha = 0$ at all points.

Or

(b) Calculate the curvature and torsion of the curve of $ax^2 + by^2 + cz^2 = 1,\ a'x^2 + b'y^2 + c'z^2 = 1$.

12. (a) Show that the osculating plane at P has in general, three point contact with the curve at P.

Or

(b) Explain : Locus of the centre of spherical curvature.

13. (a) On the paraboloid $x^2 - y^2 = z$, find the orthogonal trajectories of the sections by the planes z = constant.

Or

(b) Explain : Intrinsic properties.

14. (a) Prove that, on the general surface, a necessary condition that the curve $v = c$ be a geodesic is

$EE_2 + FE_1 - 2EF_1 = 0$, where $v = c$, for all values of u.

Or

(b) Prove that every helix on a cylinder is a geodesic.

15. (a) Derive the second fundamental form :

$Ldu^2 + 2Mdudv + Ndv^2$.

Or

(b) Prove that a necessary and sufficient condition that a curve on a surface be a line curvature is that the surface normals along the curve form a developable.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Obtain the equations of the circular helix $r = (a\cos u, a\sin u, bu), -\infty < u < \infty$ where $a > 0$, referred to s as a parameter and show that the length of one complete turn of the helix is $2\pi c$, where $c = \sqrt{(a^2 + b^2)}$.

17. Prove that the radius of curvature of the locus of the centre of curvature of a curve is given by

$$\left[\left\{\frac{\rho^2\sigma}{R^3}\frac{d}{ds}\left(\frac{\sigma\rho'}{\rho}\right) - \frac{1}{R}\right\}^2 + \frac{\rho'^2\sigma^4}{\rho^2 R^4}\right].$$

18. (a) Find a surface of revolution which is isometric with a region of right helicoid.

    (b) Explain : Isometric Correspondence.

19. (a) Prove that the curves of the family $\frac{v^3}{u^4}$ = constant are geodesics on a surface with metric

    $v^2du^2 - 2uvdudv + 2u^2dv^2 (u > 0, v > 0)$.

    (b) Explain : Geodesic curvature.

20. Derive the umbilic in the form :

$$\frac{L}{E} = \frac{M}{F} = \frac{N}{G}.$$

RW–10918

Sub. Code
511505

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

GRAPH THEORY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 mark.

1. Define a complete graph and a complete bipartite graph. Give an example of a graph which is complete as well as complete bipartite.

2. How many spanning trees are there for a tree?

3. Define Euler graph and Hamiltonian graph. Give an example of a Hamiltonian graph which is not Euler.

4. Define the closure of a graph. What is the closure of the graph K_n?

5. Is it true that every perfect matching is a maximum matching? Justify.

6. Define Ramsay number. What is $r(2,10)$?

7. Define chromatic polynomial and give an example.

8. State Brook's theorem.

9. State Euler's formula and verify it for K_4.

10. How many different directed graphs are there whose underlying graph is K_3?

Part B (5 × 5 = 25)

Answer **all** questions.

Each question carries 5 marks.

11. (a) Let T be a spanning tree of a connected graph G and let e be an edge of T. Then prove that the complement of T contains a unique bond.

Or

(b) If G is a tree, prove that $\varepsilon = v - 1$.

12. (a) Prove that a connected graph has an Euler trail if it has two vertices of odd degree.

Or

(b) Define a block. If G is a block with $v \geq 3$, prove that any two edges of G lie on a common cycle.

13. (a) Define perfect matching and prove that $K_{m,n}$ has a perfect matching if, and only if, $m = n$.

Or

(b) For any two integers $k \geq 2$ and $l \geq 2$, prove that $r(k,l) \leq r(k, l-i) + r(k-1, l)$.

14. (a) Write a note on Hajo's conjuncture.

Or

(b) Find the chromatic polynomial of K_4.

15. (a) State whether Peterson graph is planar or not. Justify.

Or

(b) Write a note on four color conjecture.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

Each question carries 10 marks.

16. Explain Kruskal's algorithm and prove that the tree constructed by the algorithm is an optimal tree.

17. If G is a simple graph with $v \geq 3$ vertices and $\delta \geq \frac{v}{2}$, then prove that G is Hamiltonian.

18. State and prove Turan's theorem.

19. State and prove Vizing theorem.

20. If D is strict and $\min\{\delta^-, \delta^+\} \geq \frac{v}{2} > 1$, then prove that D contains a directed Hamilton cycle.

RW–10919

Sub. Code
511703

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

IMAGE PROCESSING AND PATTERN RECOGNITION

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. What do you mean by gray level?

2. Define histogram

3. What is edge following?

4. Define morphology?

5. Define spatial compression?

6. What is real time image transmission?

7. What is decision function?

8. What is linear decision function?

9. What do you mean by clustering?

10. What is graph theoretic approach?

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write about Histogram equalization.

Or

(b) Explain Image processor.

12. (a) Explain about basic and second order edge detection.

Or

(b) Explain about basic morphological operation.

13. (a) Give a brief explanation of quantising compression

Or

(b) Give a brief note on image standards.

14. (a) Describe about pattern recognition.

Or

(b) Explain implementation of decision function.

15. (a) Explain about minimum distance pattern classification.

Or

(b) Explain Iso–data algorithm.

Section C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss the image representation in detail.

17. Describe the morphological operations in detail

18. Discuss image compression in detail.

19. What is decision function? Explain its methodologies.

20. Explain the pattern classification in detail.

RW–10920

Sub. Code
511101

M.Sc. (Mathematics) DEGREE EXAMINATION, APRIL 2016

ALGEBRA – I

(CBCS 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each question carries 2 marks.

1. Define normal subgroup.

2. What is the order of the product of the disjoint cycles of lengths $m_1, m_2, \ldots m_k$?

3. Show that conjugacy is an equivalence relation.

4. If A and B are groups, prove that $A \times B$ is isomorphic to $B \times A$.

5. State pigeonhole principle.

6. Define ring homomorphism.

7. State the Eisenstein criterion.

8. Define Integral domain.

9. State the fundamental theorem of R - homomorphism.

10. Prove that direct sum of free modules is a free module.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Prove that HK is a subgroup of G if and only if $HK = KH$.

Or

(b) Prove that, if ϕ is a homomorphism of G into $\overline{G}$ with kernel K, then K is a normal subgroup of G.

12. (a) Show that, if $o(G) = p^n$ where p is a prime number, then $Z(G) \neq (e)$.

Or

(b) State and prove the second part of the Sylow's theorem.

13. (a) Prove that a finite integral domain is a field.

Or

(b) Let R be a commutative ring with unit element whose only ideals are (0) and R itself. Prove that R is a field

14. (a) Prove that, if R is a unique factorization domain, then so $R[x]$.

Or

(b) Prove that if $f(x)$ and $g(x)$ are primitive polynomials, then $f(x)g(x)$ is a primitive polynomial

15. (a) Prove that the submodules of the quotient module M/N are of the form U/N, where U is a submodule of the M containing N.

Or

(b) Let R be a ring with unity. Show that an R - module M is cyclic iff $M = R/I$ for some left ideal I of R.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each questions carries 10 marks.

16. State and prove Cayley's theorem.

17. State and prove the first part of the Sylow's theorem.

18. Prove that every integral domain can be imbedded in a field.

19. State and prove the division algorithm.

20. (a) Prove that a vector space is a free module.

(b) State and prove Schur's lemma.

RW–10921

Sub. Code
511102

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

REAL ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Define : Ordered set. Illustrate.

2. Is Cantor set perfect? Why?

3. Is every uniformly continuos function continuous. Explain.

4. Do monotonic functions have discontinuities of the second kind.

5. State the mean value theorem.

6. If f is differentiable on $[a,b]$, the f' can have discontinuties on $[a,b]$. Your comment.

7. State the theorem on integration by parts.

8. Define : upper and lower Riemann integrals.

9. Give a criterion for uniform convergence.

10. Define : Equicontinuity.

Section B $(5 \times 5 = 25)$

Answer **all** the questions.

Each question carries 5 marks.

11. (a) State and Prove the archimedean property of R.

Or

(b) State and Prove the Heine – Bovel theorem.

12. (a) Let E be a noncompact set in R'. Show that there is a continuous and bounded function on E which has no maximum.

Or

(b) Let f be a continuous real function on a metrix space X. Prove that the set of all $p \in X$ at which $f(p) - 0$ is closed.

13. (a) If f has a local maximum at a point $x \in (a,b)$ and if $f'(x)$ exists, then show that $f'(x) = 0$.

Or

(b) Let f be defined for all real x, and $|f(x) - f(y)| \leq (x-y)^2$ for all real x and y. Prove that f is constant.

14. (a) If f is continuous on $[a,b]$ then f is Riemann integrable on $[a,b]$. Justify.

Or

(b) State and Prove the fundamental theorem of calculus.

15. (a) State and Prove weierstrass test for uniform convergence.

Or

(b) If a sequence of continuous functions $\{f_n\}$ converges uniformly on a compact k, then prove that $\{f_n\}$ is equicontinuous on k.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that every interval $[a,b]$ $(a<b)$ is uncountable.

17. Prove that a continuous real function assumes all intermediate values on an interval.

18. State and prove Taylor's theorem.

19. Prove that the length of a rectifiable curve can be given by a Riemann integral.

20. State and prove the stone – Weierstrass theorem.

RW–10922

Sub. Code
511103

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

DIFFERENTIAL EQUATIONS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Gramian of continuous functions $\varphi_n(x),\ldots\ \varphi_n(x)$ on $a \le x \le b$.

2. Find a polynomial solution of $y'' - 2xy' + 4y = 0$.

3. Find all solutions of $2x^2y'' + xy' - y = 0$, $x > 0$.

4. What are the singular points of the hypergeometric equation?

5. Find a complete integral of the equation $(p+q)(z - xp - yq) = 1$.

6. Explain : Integral surface, Integral curves.

7. State Poisson's equation. Give the physical meaning.

8. Classify the equation $\frac{\partial^2 z}{\partial x^2} + x\frac{\partial^2 z}{\partial y^2} = 0$.

9. State exterior Neumann problem for Laplace equation.

10. State Helmholtz's equation.

Part B (5 × 5 = 25)

Answer **all** questions.

Each question carries 5 marks.

11. (a) One solution of $y'' + \frac{1}{4x^2} y = 0$ is $\varphi(x) = x^{1/2}$, $x > 0$. Find the another solution.

Or

(b) Find all solutions of the equation $y'' - \frac{2}{x^2} y = x$, $0 < x < \infty$.

12. (a) Find all solutions of the equation $x^2 y'' - 5xy' + 9y = x^3$, for $x > 0$.

Or

(b) Show that infinity is a regular singular point for the Legendre equation.

13. (a) State Sonia Kowalewski theorem.

Or

(b) Find the general integral of the equation

$$z\left(x\frac{\partial z}{\partial x} - y\frac{\partial z}{\partial y}\right) = y^2 - x^2.$$

14. (a) Reduce the equation $\frac{\partial^2 z}{\partial x^2} = x^2 \frac{\partial^2 z}{\partial y^2}$ to canonical form.

Or

(b) Solve : $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$ subject to $z(0, t) = 0$ and $z(a,t) = 0$ for all t.

15. (a) Establish a necessary condition for the existence of the solution of the interior Neumann problem.

Or

(b) Obtain d'Alembert solution of the one-dimensional wave equation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that Legendre polynomials are orthogonal on $[-1.1]$.

17. Solve the Bessel equation of order α.

18. Find the integral surface of the equation

$$2y(x-3)\frac{\partial z}{\partial x} + (2x - z)\frac{\partial z}{\partial y} = y(2x-3)$$

Which passes through the circle $z = 0$, $x^2 + y^2 = 2x$.

19. Find the solution of $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=e^{-x}\cos y$ which tends to zero as $x \to \infty$ and has the value $\cos y$ when $x=0$.

20. A rigid sphere of radius a is placed in a stream of fluid whose velocity in the undisturbed state is V. Determine the velocity of the fluid at any point of the disturbed stream.

RW–10923

Sub. Code
511501

M.Sc. DEGREE EXAMINATION, APRIL 2016

First Semester

Mathematics

ANALYTIC NUMBER THEORY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. If $(a, b) = 1$, then prove that $(a^n, b^k) = 1$ for all $n \geq 1, k \geq 1$.

2. If n is a composite number, is it true that $(n, \phi(n)) > 1$? Justify.

3. If $x \geq 2$, then prove that $\log[x]! = x \log x - x + 0(\log x)$.

4. If $x = k + y$ where k is an integer and $0 \leq y < 1$, then prove that $k = [x]$.

5. State Abel's identity.

6. Write down Merten's conjecture.

7. If $a \equiv b(\text{mod } m)$ and if d/m and d/a then prove that d/b.

8. Solve the convergence $25x = 15 \pmod{120}$.

9. State gauss lemma.

10. Find the value of $(2/13)$ and $(2/7)$.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Given any two integers a and b, show that there is a common divisor d of a and b of the form $d = ax + by$, where x and y are integers.

Or

(b) State and prove Mobius inversion formula.

12. (a) Show that the set of lattice points visible from the origin has density $6/\pi^2$.

Or

(b) For all $x \geq 1$ prove that $\left|\sum_{n \leq x} \Sigma \frac{\mu(n)}{n}\right| \leq 1$, with equality holds only if $x < 2$.

13. (a) State and prove the Abel's identity.

Or

(b) Show that $\lim_{x \to \infty}\left(\frac{M(x)}{x} - \frac{H(x)}{x \log x}\right) = 0$.

14. (a) Show that the convergence is an equivalence relation.

Or

(b) State and prove Wolstenholme's theorem.

15. (a) State and prove Euler's criterion.

Or

(b) State and prove reciprocity law for Jacobi symbols.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. For $n \geq 1$, prove that $\phi(n) = n \prod_{p/n}\left(1 - \frac{1}{p}\right)$ and state and five properties of ϕ and prove the same.

17. (a) State and prove the Euler's summation formula.

(b) State and prove Legendre's identity.

18. Let p_n denote the nth prime. Prove that the following asymptotic relations are logically equivalent

(a) $\lim_{x\to\infty} \frac{\pi(x)\log(x)}{x} = 1$

(b) $\lim_{x\to\infty} \frac{\pi(x)\log \pi(x)}{x} = 1$

(c) $\lim_{x\to\infty} \frac{p_n}{n\log n} = 1$.

19. (a) State and prove Lagrange's theorem.

(b) State and prove Little Fermat theorem.

20. State and prove the quadratic reciprocity law.

RW–10924

Sub. Code
511201

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

ALGEBRA — II

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define an inner product space with an example.

2. Define an annihilator with an example.

3. Define an algebraic number with an example.

4. Prove $x^3 - 1$ is irreducible over the field of rational number.

5. Define the fixed field with an example.

6. Define solvable group with an example.

7. Define the invertible transformation with an example.

8. Define the characteristic root.

9. Define the symmetric matrix with an example.

10. Define the Hermitian adjoint.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Prove that the intersection of two subspaces of V is a subspace of V.

Or

(b) State and prove Schwarz inequality.

12. (a) Prove that a polynomial of degree n over a field can have at most n roots in any extension field.

Or

(b) Prove that any finite extension of a field of characteristic 0 is a simple extension.

13. (a) Prove that the fixed field of G is a subfield of K.

Or

(b) Prove that S_n is not solvable for $n \geq 5$.

14. (a) If $\lambda_1, \ldots, \lambda_k$ in F are distinct characteristic roots of $T \in A(V)$ and if $u_1, \ldots, u_k$ are characteristic vectors of T belonging to $\lambda_1, \ldots, \lambda_k$ respectively, show that $u_1, \ldots, u_k$ are linearly independent over F.

Or

(b) Prove that two nilpotent linear transformations are similar if and only if they have the same invariants.

15. (a) For $A, B \in F_n$ and $\lambda \in F$, prove that $tr(AB) = tr(BA)$.

Or

(b) Prove that any two finite fields having the same number of elements are isomorphic.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. State and prove Gram-Schmidt orthogonalization process.

17. Prove that the element $a \in K$ is algebraic over F if and only if $F(a)$ is a finite extension of F.

18. State and prove the fundamental theorem of Galois theory.

19. If $T \in A(V)$ has all its characteristic roots in F, show that there is a basis of V in which the matrix of T is triangular.

20. Prove that the linear transformation T on V is unitary if and only if it takes an orthonormal basis of V into an orthonormal basis of V.

RW–10925

Sub. Code
511202

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

MEASURE AND INTEGRATION

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define Lebesgue outer measures.

2. Define Lebesgue measurable.

3. Define Lebesgue measurable function.

4. Prove that the continuous functions are measurable.

5. Define Lebesgue integral.

6. Show that $\int_1^\infty \frac{dx}{x} = \infty$.

7. Define singed measure.

8. Show that the signed measure v is finite or σ-finite respectively if, and only if, $|v|$ is, or if, and only if, both v^+ and v^- are.

9. Define measurable rectangle.

10. Show that if $\{A_i\}$ is a monotone square of sets, then $LimA_i^y = (Lim\ A_i)^y$ and $Lim(A_i)_x = (Lim\ A_i)_x$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that if μ is a σ -finite measure on $\Re$, then the extension $\overline{\mu}$ of μ to S^* is also σ–finite.

Or

(b) Prove that every interval is measurable.

12. (a) Prove that the following statements are equivalent.

(i) f is measurable function

(ii) $\forall \alpha, [x : f(x) \geq \alpha]$ is measurable

(iii) $\forall \alpha, [x : f(x) < \alpha]$ is measurable

(iv) $\forall \alpha, [x : f(x) \leq \alpha]$ is measurable.

Or

(b) Prove that there exists a non - measurable set.

13. (a) State and prove Lebseugue's monotone convergence theorem.

Or

(b) Show that $Lim \int \frac{dx}{\left(1+\frac{x}{n}\right)^n x^{\frac{1}{n}}} = 1$.

14. (a) Prove that a countable union of sets positive with respect to a singed measure v is a positive set.

Or

(b) State and prove Jordan decomposition.

15. (a) State and prove Fubini's theorem.

Or

(b) Prove that ξ is algebra.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that the outer measures of an interval equal its length.

17. Let c be any real number of let f and g be real-valued measurable functions defined on the same measurable set E. Then prove that $f+c, cf, f+g$ and fg are also measurable.

18. State and prove Lebesgue's dominated convergence theorem.

19. State and prove Radon-Nikodym theorem.

20. Let f be a non-negative $S \times \Im$ – measurable function and let $\phi(x) = \int_X f_x \, dv, \psi(y) = \int_X f^y d\mu$ for each $x \in X, y \in Y$, then prove that ϕ is S- measurable, ψ is a $\Im$-measurable and $\int_X \phi \, d\mu = \int_{X \times Y} f \, d(\mu \times v) = \int_Y \psi \, dv$.

RW–10926

Sub. Code
511203

M.Sc. DEGREE EXAMINATION, APRIL 2016

Mathematics

COMPLEX ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. If $g(w)$ and $f(z)$ are analytic functions, then prove that $g(f(z))$ is also analytic.

2. Using Hadamard's formula, find the radius of convergence of $\sum n^p z^n$.

3. Compute $\int_{|z|=1} e^z z^{-n} dz$.

4. Define Removable singularity. Give an example.

5. How many roots of the equation $x^4 - 6z + 3 = 0$ have their modulus between 1 and 2?

6. Find the residue of $f(z) = \frac{1}{(z^2-1)^2}$ at each of its poles.

7. Prove that the arithmetic mean of a harmonic function over concentric circles $|z| = r$ is a linear function of $\log r$.

8. Develop $\frac{1}{(1+z^2)}$ in powers of $(z-2)$.

9. State Mittag – Leffler's theorem.

10. Define genus of a canonical product.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Is an analytic function $f(z)$ with a constant modulus reduces to a constant? If yes, Justify your answer.

Or

(b) State and prove the Symmetry principle theorem.

12. (a) State and prove Cauchy's integral formula.

Or

(b) Suppose that $f(z)$ is analytic in the region Ω' obtained by omitting a point a from the region Ω. Prove that a necessary and sufficient condition that there exist an analytic function in Ω. Which coincides with $f(z)$ in Ω' is that $\lim_{z \to a}(z-a)f(z)=0$.

13. (a) State and prove Rouche's theorem.

Or

(b) Evaluate $\int_0^\infty \frac{x \sin x}{x^2+a^2} dx, (\text{a real})$.

14. (a) Prove the Mean value property for harmonic functions.

Or

(b) State and prove Weierstress's theorem.

15. (a) With the usual notations, prove that

$$\frac{\pi^2}{\sin^2 \pi z} = \sum_{n=-\infty}^{\infty} \frac{1}{(z-n)^2}.$$

Or

(b) Derive the Jensen's formula.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. State and Prove Abel's theorem.

17. State and Prove the Cauchy's theorem for a rectangle.

18. State and Prove general Cauchy's theorem.

19. Obtain the Laurent expansion $\sum_{n=-\infty}^{\infty} An(z-a)^n$ for the function $f(z)$ analytic in $R_1 < |z-a| < R_2$.

20. (a) With the usual notations, prove that $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$.

(b) Prove that for $|z|<1, (1+z)(1+z^2)(1+z^4)(1+z^8)\cdots = \frac{1}{1-z}$.

RW–10886

Sub. Code
511201

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Mathematics

ALGEBRA – II

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define inner product space.

2. In a vector space show that $\alpha(v-w)=\alpha v-\alpha w$.

3. Prove that $F(a,b)=F(b,a)$.

4. Define simple extension.

5. Define solvable group.

6. Define fixed field.

7. Define invertible.

8. In F_2 prove that for any two elements A and B, $(AB-BA)^2$ is a scalar matrix.

9. Define symmetric matrix and skew symmetric matrix.

10. State Jacobson lemma.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Let V be the internal direct sum of $U_1, \ldots\ldots U_n$. Prove that V is isomorphic to the external direct sum of $U_1, \ldots\ldots U_n$.

Or

(b) If V is finite dimensional over F then prove that any two bases of V have the same number of elements.

12. (a) State and prove Reminder theorem.

Or

(b) Prove that the real number α is constructible if and only if we can find a finite number of real numbers $\lambda_1, \ldots\ldots \lambda_n$ such that

(i) $\lambda_1^2 \in F_0$

(ii) $\lambda_1^2 \in F_0(\lambda_1, \ldots\ldots\ldots \lambda_{i-1})$ for $i = 1, 2, \ldots\ldots n$ such that $\alpha \in F_0(\lambda_1, \ldots\ldots\ldots \lambda_n)$.

13. (a) Prove that the fixed field of G is a subfield of K.

Or

(b) Prove that K is a normal extension of F iff K is the splitting field of some polynomial over F.

14. (a) Let V be a finite dimensional vector space over F. Let $S, T \in A(v)$. Prove that

(i) $r(ST) \leq r(T)$

(ii) $r(TS) \leq r(T)$ and

(iii) $r(ST) \leq r(TS) = r(T)$, for S regular in $A(V)$.

Or

(b) If $T \in A(v)$ is nilpotent, then prove that $\alpha_0 I + \alpha_1 T + ...\alpha_m T^m$, where the $\alpha_1 \in F$, is invertible if $\alpha_0 \neq 0$.

15. (a) Prove that for $A, B \in F_n$ and $\lambda \in F$:

(i) $tr(\lambda A) = \lambda tr A$

(ii) $tr(A + B) = tr A + tr B$

(iii) $tr(AB) = tr(BA)$.

Or

(b) If two rows of A are equal then prove that $\det A = 0$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If V and W are of dimensions m and n respectively, over F, then prove that $Hom(V, W)$ is of dimension mn over F.

17. Prove that the element $a \in K$ is algebraic over F if and only if $F(a)$ is a finite extension of F.

18. If $p(x) \in F[x]$ is solvable by radicals over F, then prove that the Galois group over F of $p(x)$ is a solvable group.

19. Prove that there exists a subspace W of V, invariant under T, such that $V = V_1 \oplus W$.

20. (a) If $T \in A(V)$ then prove that $T^* \in A(V)$. Moreover,

    (i) $(T^*)^* = T$

    (ii) $(S+T)^* = S^* + T^*$

    (iii) $(\lambda S)^* = \bar{\lambda} S^*$

    (iv) $(ST)^* = T^* S^*$, for all $S, T \in A(V)$ and all $\lambda \in F$.

    (b) $T \in A(V)$ is unitary iff $TT^* = 1$.